QUANTITATIVE CATEGORIZATION OF TONAL MUSIC STYLES

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ABSTRACT

Tonal music style is a substantial quality of the music material that can be recognized to belong to individual composers or their epoch. We were interested in the aspects of style that are detectable in notation. The objective of this study was to identify the main variables that allow for effective classification of tonal music styles. This was a quantitative study based only on measurable features giving rise to numerical continuous variables resulting from actual measurements carried out by software. As many features as practicable were operationalized and software routines devised to carry out the measurements on a purposely built database that stood for the whole range of styles of the period of common practice. The resulting dataset was investigated by machine learning algorithms. Key and rhythm measures and frequencies of use of scale degrees turned out to be the variables of highest predictive validity. The three key indicators were then used to effectively classify pieces by composer.

1. INTRODUCTION

Paisley [1] is credited with the concept that style is best accounted for by ``minor encoding habits''. He quoted Bernard Berenson's statement [1:221]: ``Obviously what distinguishes one artist from another are the characteristics he does not share with others'' leading to the notion of style as `deviation from the norms'. This idea opens the door for quantitative analyses of music. Although there is a long tradition of these, going back to Gabura [2][3], they continue to be in the minority. Recently, Clarke & Cook [4] have provided support for this approach which ``embodies a principled awareness of both the potential to engage with large bodies of data, and the appropriate methods for achieving this''. This converges with David Huron's [5] statement that ``quantitative methods allow us to observe patterns that might otherwise be difficult or impossible to decipher''.

Within this trend, our own interest points specifically to style, a concept whose use is ubiquitous but a phenomenon that is neglected in itself. For example, Klaus Döge, discussing Dvorák [6] characterizes his American period by the use of ``pentatonism in the melodic line, a flattened leading note, plagal cadences, drone accompaniment, rhythmic ostinato and strongly syncopated rhythms''. Such list may make us feel impressed by the acumen of the analyst, but it may also bring about a lingering uneasiness caused by the lack of a conceptual frame of reference for style. The reader could wonder whether such tally is really exhaustive, or is it perhaps biased, and what is the possible range of the list of features from which the analyst picked these items. Naturally, these doubts would not have been raised if there was an established conceptual structure for style, which makes one wonder how it is possible that a fully developed theoretical characterization of tonal music style does not yet exist. It is likely that the frequent encounter with multiple works of a certain composer leads our minds to the formation of a certain `prototype' of the composer's style, to the point that we can roughly classify those works in relation to how representative or idiomatic they are. But this process is entirely unsystematic, and its limitations become apparent when one is faced with the discovery of a similar work of unknown composer. There are no tools to measure how close the new work is to others of the probable composer to justify attribution.

The notion of developing tools for attribution based on quantitative measures led our interest to the aspects of style that are detectable in notation, which is all we have when a score is discovered. If a system of fundamental measurable variables could be found that characterized style on the basis of notation alone, the basis for the development of attribution tools would have been set. The quantitative approach is ideally suited to this pursuit. This study is based on the notion that a comprehensive enough set of features that allows for stylistic classification of musical works should lead to the identification of the main conceptual dimensions of style. Measuring as many meaningful parameters of the music as possible would result in a set of values amenable to quantitative analyses. The goal involved avoiding subjective judgments, and for this purpose it would be ideal to preferably rely on continuous variables resulting from actual measurements carried out by purposely-designed algorithms implemented as software.

2. METHOD AND MATERIALS

An overview of the method comprises operationalizing as many features as practicable; then devising algorithms to measure them and implementing them into software; using these programs on a suitable sample of pieces standing for the period of common practice; and finally analyzing the resulting data. In this process, three stages can be recognized:

2.1.1. Corpus collection and digitization}

The need of a suitable sample of the whole range of styles of the period of common practice on one side, and the time limits imposed by practical constraints on the other, led to the compromise of creating a database of 297 piano or harpsichord pieces, i.e., 11 pieces each from from 27 leading composers from Handel to Shostakovich. The pieces were selected by size and assumed representativeness, scanned and OCR'ed into musicXML files.

2.1.2. Feature selection and measurement

Observable features were extracted from the main areas of music, namely key, rhythm, harmony, and melody. Each feature was operationalized and software algorithms were developed to measure them preferably into continuous numerical variables. The algorithms were applied to the database, thus yielding a set of numerical results for each piece.

2.1.3. Data analysis and interpretation

These results were collected in spreadsheets to make them available to analytical algorithmic methods in order to identify the main dimensions. Once these were identified, they were used to classify the database by composer.

2.2. Key determination

All the features, except for the rhythm variables, are pitch-related. Therefore, an essential initial step was the automatic determination of key as a point function, i.e. at every point of the score.

This task was carried out according to a method described in [7], which is essentially based on finding the maximum dot-product between the pitch content of the piece and a set of numbers statistically obtained from a large set of works which represent the ideal frequencies of the scale degrees. The fact that, in major and minor scales based on the same tonic, the so-called modal degrees fall on different notes, made necessary to process everything separately by mode. The set of numbers used is shown in Table 1.

Table 1 - Key set for both modes

2.3. Sliding window

Scale degree	Major	Scale degree	minor
Tonic	16.80	Tonic	18.16
Raised Tonic	0.86	Lowered Supt.	0.69
Supertonic	12.95	Supertonic	12.99
Raised Supt.	1.41	Mediant	13.34
Mediant	13.49	Lowered Subd.	1.07
Subdominant	11.93	Subdominant	11.15
Raised Subd.	1.25	Lowered Dom.	1.38
Dominant	20.28	Dominant	21.07
Raised Dom.	1.80	Submediant	7.49
Submediant	8.04	Lowered Subt.	1.53
Raised Subm.	0.62	Subtonic	0.92
Subtonic	10.57	Lowered Tonic	10.21

This method determines the key of a fragment of the piece that can stretch to the whole of the piece, giving in that case the global key. But practically every piece modulates at some point, which introduces a confounding variable if the initial key is assumed throughout. The way to obtain the key for each section as well as the points of modulations is to calculate it during a window wide enough to be stable and short enough not to bypass modulating passages and then slide the window along the score, repeating the calculation as new notes are being added and the old ones dropped.

While the key remains the same, successive calculations yield results that vary slightly. If the piece modulates, the key can be seen as a piecewise function of time affected by noise. The window is a low-pass filter whose resolution and stability depend on its width. The solution is a compromise.

2.4. Measured features

The following are the features that were measured in each of the music areas:

2.4.1. Key

In order to assess key parameters, three scalar variables were measured:

(a) A previous study [8] had suggested that the tonal weakness of a piece correlates with the rate of decay of the dot products, that is, the less clearly tonal a piece is, the more slowly falls the value of the successive dot products. This led to the calculation of the average percental distance between the top scalar dot product for each slice and each of the five runners-up. The mean of those averages for the whole of the piece was labeled *Global average dot product*. It was found that historically, its value is relatively stable before 1800 but declines gradually from then on until the eve of the Great War. Its absolute value, which is affected by parameters such as the width of the measuring window, has not been normalized but with the parameters used in this study, its values for our database varied from a minimum of

560 (Prokofiev: March Op.3 No.3) to a maximum of 2022 (Satie: Gnossienne No.1).

(b) An index that indicates the propensity of the piece to modulate chromatically was called *Modulation index*, and its value calculated as the percental ratio between the sum of the tonal distances for each key transition and the normalized total duration of the piece. For our database, the values of this index varied between a minimum of 0 and a maximum of 30.662 (Prokofiev: Visions fugitives Op.22 No.5)

(c) Likewise, a similar index for the propensity of the piece to modulate diatonically was called *Intermodal index* and calculated as the percental ratio between the sum of the inter-mode changes and the normalized total duration of the piece. For our database, the values of this index ranged between a minimum of 0 and a maximum of 7.033 (Shostakovich Prelude Op.34 No.5).

2.4.2. Rhythm

As rhythm is a multi-dimensional phenomenon, a rhythm-measuring scheme was devised comprising four practically independent scalar variables:

(a) *Beat rate*, measured as beats per minute. For our database, the beat rate varied between a minimum of 24 (Satie: Gnossienne No.5) and a maximum of 510 (Scriabin: Etude Op.8 No.3);

(b) Number of *onsets per measure*. For our database the value of this variable ranged between 5.29 (Liszt: Nuages gris) and 52.245 (Ravel: Ondine);

plus two indexes of rhythmic variability that refer to the pattern of onsets for the piece:

(c) A way to measure changes in the longitudinal profile of the piece would be to consider the variations in the number of onsets taking place at each subdivision of the bar. Thus, the Index of longitudinal or *Between-measures* variability, is equal to the ratio of the sum of the variances for the number of onsets occurring at each subdivision of the measure and the total number of onsets. For our database, the value of the Between-measure variability varied between a minimum of 0.144 (J.S.Bach: Gigue from French Suite No.3) and a maximum of 24.953 (Debussy: Prelude No.1); And

(d) Likewise, a way to capture the variability occurring within each measure would entail considering the variations across all the subdivisions of each bar. Thus, the Index of transversal or *Within-measures* variability, equal to the ratio of the sum of the variances for the number of onsets occurring in each measure and the total number of onsets. For our database, the value of the Within-measure variability ranged between a minimum of 0.239 (Chopin: Etude Op.25 No.12) and a maximum of 24.953 (Ravel: Menuet from Sonatine).

2.4.3. Scale degrees

One of the simplest measurements that can be effected, the frequencies of use of the individual twelve scale degrees - in both modes - was carried out on each piece. Since each piece yields 24 values, these can be considered a set of 24 scalar variables, or more properly a single 24-component vectorial variable.

2.4.4. Harmony

Using the concept of sonority [9] rather than chord to refer to each simultaneous combination of pitch classes, what was initially measured - separately for major and minor modes - was the frequency of use of each of the possible sonorities made of up to four different notes, i.e. the single monad, the six dyads, the 19 triads and the 43 tetrads. These values, however, constitute a sparse matrix because the majority of the sonorities are seldom used. A double set of 69 scalar variables, the majority of which is zero, was not very convenient for analysis. The consideration that only a few of the sonorities receive a name in harmonic theory - the six dyads, five triads and seven tetrads - suggested that it was better to limit the measurements to this 18-variable reduced set.

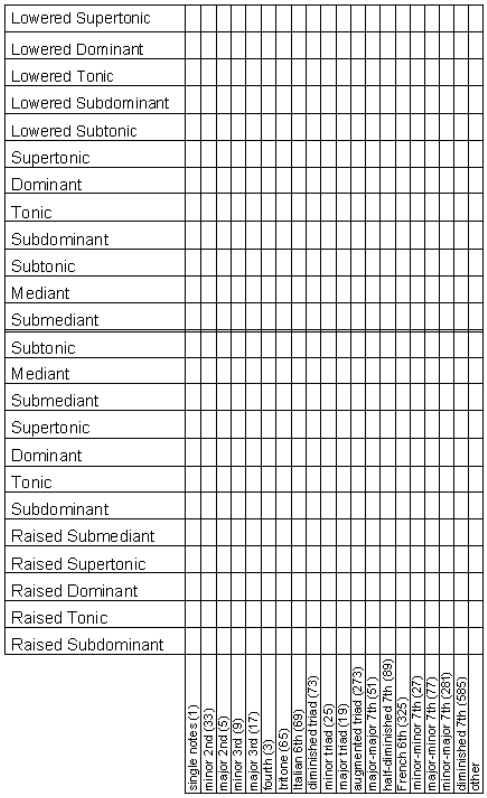


Fig.1 - Grid for representation of harmonies

Considering, for example, that each sonority, such as major and minor triads and major-minor sevenths, can play different harmonic functions such as tonic, dominant, and so on, it is arguable that the frequency of use of each combination of sonority and function will have stylistic significance. The concept can be generalized to the set of all of the sonorities and all the

functions, resulting in the measurement of the frequency of use of each of the sonorities playing each of the harmonic functions. Fig.1 illustrates the grid comprising the combinations of sonorities and functions, and Fig.2 shows an example, the frequency of the sonority-function combinations for Handel's Sarabande from Suite 16.

This set turns out to comprise 432 scalar variables, although it can be more properly viewed as a single second order tensorial variable for each piece in the database. This compares unfavorably with the size of the database, which comprises only 297 cases, and is also a sparse matrix, as Fig.2 illustrates - for most of the squares the frequency is zero. Consequently, we decided to collapse the matrix in two directions, thus replacing the 432-component second order tensor by two vectorial variables, one an 18-component measure of harmonic frequencies (irrespective of function), and the other a 24-component measure of the frequencies of harmonic functions (irrespective of harmony).

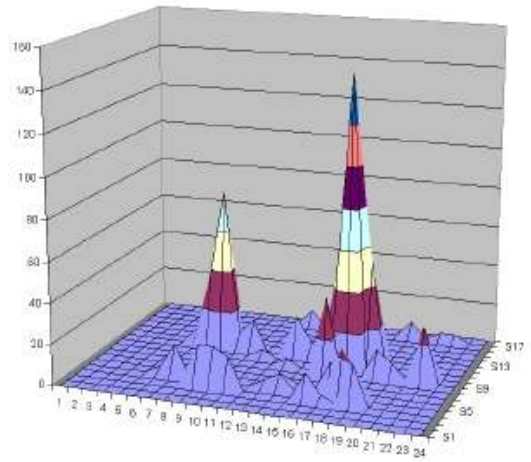


Fig.2 - Harmony frequencies for Handel's Sarabande from Suite 16

2.4.5. Melody

In keyboard music, in general, voices cannot be automatically individualized. Particularly after the end of the Classical period, `voices' exist only in the mind of the composer and they tend to appear and vanish as a matter of convenience. As counterpoint lines are beyond disentangling, it was thought necessary to approach the materials as if they were basically melodies accompanied by a bass line. Unfortunately, the perception of melody is linked to psychoacoustic principles and there does not seem to be any valid strategy to objectively identify it. This led to the conclusion that the only practical way to operationalize a melody-like variable was to consider the frequency of use of scale degrees occupying the highest pitch in the piece. This operationalization in many cases will coincide with the melody, in others it will not, but it has the advantage of measuring an objective variable.

Consideration of the two-part principle suggested also the convenience of carrying out a similar measure for the bottom note of the music, which would agree with Schoenberg's [10] referring to the bass as ``the second melody'' in his theoretical writings. Therefore, the melody-related variables were two 24-component vectorial variables measuring the scale degree frequencies of the notes occupying the highest and lowest positions at each point in the piece.

3. RESULTS

3.1. Data

The preceding feature measurements were carried out by software routines applied to each of the pieces in the database. The result was a set of 139 numerical continuous variables which constitute the dataset for the analyses. As the expectation was to use statistical techniques for the analyses, data were subjected to the customary screening for distribution and subsequent transformation of variables when they did not comply with acceptable levels of normality and skewness.

It has to be remembered that the main purpose of the study was to unveil the main conceptual dimensions of style, and the expectation, that measuring a comprehensive set of features would capture the style of the pieces. According to this goal, it is necessary to analyze the whole of the dataset together. For example, if the most important dimensions were to be identified, they would have to emerge as the principal components of the complete set of variables.

Nevertheless, the dataset proved impregnable for statistics. The number of variables was too large for any statistical technique except for Principal Component analysis. This would have been the appropriate technique for the stated purpose, but Principal Component Analysis can be applied only when there are clusters of variables in the data. This is assessed by mens of the the values of the covariance matrix as well as the Kaiser-Meyer-Olkin measure of sampling adequacy. Both were found to be below acceptable levels, meaning that Principal Components was not an appropriate way to analyze the data because there were no significant factors to be extracted.

3.2. Data mining algorithms

Machine learning algorithms, being assumption-free, non-linear and non-parametric and some - like Classification and Regression trees (CART) and Random Forests - being able to handle highly complex data structures with a large number of variables such as this dataset, constitute a more powerful alternative to statistics, and were the tool of choice for the analyses.

3.2.1. CART

The CART algorithm was developed by Breiman, Friedman, Olshen and Stone [11]. A complete

mathematical description of the algorithm is given in Breiman et.al, (1984). Readable introductions are available in Hastie, Tibshirani and Friedman (2009) and Steinberg (2009). The CART algorithm represented a major milestone in the evolution of artificial intelligence [12]. The CART unsupervised machine learning procedure, used when there are no target variables, is an ideal alternative to traditional clustering algorithms [13]. CART unsupervised learning was used to identify two or three important variables that would be responsible for partitioning the data into homogeneous segments or clusters. This procedure lets the whole set of predictor values determine the model without need for variable transformations. Starting on node 1 containing all the cases, the first binary split is done using *modulationindex*. If the value of this variable is < 4.01, cases are separated into the left child node; if the value is > 4.01 they are separated into the right child node. Further binary splitting then occurs on the children nodes. The variable used as the binary splitting variable on both of these child nodes *intermodalindex*. Unsupervised CART, unlike a traditional CART model, does not require an optimal sized tree to be developed. The tree model in unsupervised CART can be pruned back as required to reveal data groupings of interest or significance. In our analysis, pruning the tree back to the first two binary splits provides an important insight about the two variables that are the most important in the initial tree splits.

3.2.2. Random Forests

CART introduced the concept of a tree model. Random Forests extended this concept by developing a collection of individual trees or a ``tree ensemble''. Although each tree is independent of any other, the results of each of these trees is combined via a voting process (for categorical models) or by averaging (for regression models). Since each tree is independent, adding trees to the model does not create overfitting. In this way, this algorithm tends to washout any artifacts and model instabilities that can occur with single tree models, yielding results that are robust and stable [14]. A complete mathematical description of this algorithm is also given in this source.

Random Forests was run to classify the database, with composers being the target, or *Y*, variable. After the run, the algorithm provides the list of overall variables of importance for the classification. The list shows that:

- the most important three variables are the three key indicators:
  - *modulation index*
  - *intermodal index*
  - *Global average dot product*
- of the next five variables, four are the rhythm indicators;
- with the exception of two functions and one harmony, the remaining 24 variables are the frequencies of use of all the scale degrees.

Thus, from a conceptual point of view, Random Forests classifies fundamentally on the basis of the key, rhythm and scale degree variables.

3.2.3. Gene Expression Programming (GEP)

The previous two algorithms agreed about the most important variables that can be used for a classification model (*Y* = the composer). In view of those results, the three key variables found to be of the greatest importance from the CART and Random Forests models were included within a new evolutionary algorithm called Gene Expression Programming (GEP). GEP was developed by C. Ferreira [15] and can be used to construct binary classification models or regression models. *Y* takes the value of 1 for a particular composer, 0 otherwise; there being 27 different composers, 27 binary GEP classification models were run. Using standard GEP binary classification settings gave an overall classification accuracy of 86.6% with a minimum of 74.1% for Scriabin and a maximum of 95.6% for Schubert.

To remove any suspicion that GEP could have overfitted the classification solutions, a second trial was carried out, splitting the 297 dataset records into a training set of 189 records, and an independent test set of 108 records. This trial was run on six composers chosen at random. In this trial the GEP models gave a classification accuracy for the training set of 80.9%. The average classification accuracy for the test set was 72.8%. As expected there is a deterioration for the training values due to the reduction in size of the training dataset. Likewise, the test values are lower since validation is done using only 36% of the original dataset records; however, being able to achieve a predictive accuracy across the six randomly selected composers of over 72% on the basis of completely new pieces, is a remarkable result.

4. DISCUSION

The limitations of statistics have been made explicit by their inability to deal with the dataset of this study. Two independent data mining algorithms, however, have shown their effectiveness and the agreement of their results. The question may arise whether, once the main dimensions for classification have been identified, would it not be possible to do statistical studies based on these variables? In our view, the central problem is that the dataset of this study turned out to be very complex in the sense of comprising non-linear variables with complex interactions. Data mining algorithms can deal with a large number of variables without imposing conditions on their distributions, linearity of mutual interactions, but statistic methods cannot. Nevertheless, two attempts were carried out to see if it was possible to cluster the

data through statistical methods using the found most important variables as predictors. The SPSS Two-step cluster is a method supposed to automatically find the optimum number of clusters. Two-step was tried using the three key variables as predictors, but the result was that the procedure identified a single cluster comprising the whole dataset. A second attempt used as predictors the best 13 variables as identified by Random Forests. But for a second time, Two-step cluster put all the cases in a single cluster.

Therefore, the knowledge of the most important variables of the dataset does not bring the classification of tonal music within the reach of statistics. But since powerful data mining algorithms are available for the home computer, studies of this kind could become more widespread to the extent that the data mining software becomes affordable for the public.

5. CONCLUSIONS

This study has identified, among the measured features, the most important dimensions of tonal music style. These are measures of key and mode changes, rhythm measures and the relative frequency of use of scale degrees. These could be the basis of a conceptual model and a basic taxonomy of tonal music styles. Naturally, it is not possible to exclude the existence of other unmeasured variables that could turn out to be of comparable importance. The first candidates could be the frequencies of transitions between scale degrees and between harmonies. But notice that 24 chromatic scale degrees give rise to 552 possible transitions between them, a set of variables that far exceeds the total number of variables used in this study. Considering transitions probably requires a different approach.

The study has also proven it is possible to automatically identify a composer's style from the score alone, and created a system for the identification, which is a valuable contribution to authorship studies.

We believe this study opens new grounds for musical research both in philosophical and methodological senses. It shows it is possible to take a quantitative approach, generating quantitative measures for music variables and carrying out precise measurements, excluding as much as possible the influence of subjective judgments. It also shows the value of classifying and clustering music scores (and composers) by data mining methods as opposed to outdated statistical procedures.

Also, on the methodological side, the understanding that the tonalness of music is based on highly consistent proportions of use of the scale degrees makes indefensible for musical studies to be based on notes instead of scale degrees, and suggests the necessity of adopting a reliable method to determine key as a point function.

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